PROJECT BASED LEARNING (PBL) REPORT

ON

DESIGN OF A FUZZY CONTROLLER FOR CAMERA AUTOFOCUS CONTROLLER

Bachelor of Engineering

Submitted by

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1. Introduction

Autofocus (AF) is a mechanism used in modern digital cameras to automatically bring the subject of an image into sharp focus without manual adjustment of the lens by the user. Conventional autofocus controllers rely on precise mathematical models relating lens position to image sharpness; however, the relationship between focus error, scene contrast, and lens movement is inherently non-linear, noisy and difficult to model exactly. Fuzzy logic control (FLC) provides a robust alternative, as it can incorporate human-like heuristic reasoning (expressed as linguistic IF-THEN rules) to drive the lens motor smoothly and quickly towards the point of best focus, even in the presence of uncertainty and imprecise sensor data.

This report presents the design of a two-input, single-output Fuzzy Logic Controller (FLC) for a camera autofocus system. The controller observes the current sharpness error and its rate of change, and produces a crisp control signal that drives the lens focusing motor. The design follows the standard fuzzy controller development procedure: identification of inputs/outputs, definition of linguistic variables and membership functions, construction of the rule base, and fuzzy inference with defuzzification using the weighted average method.

2. Objective

- To identify the input and output variables of a camera autofocus controller.
- To define the range and linguistic variables/values for each input and output.
- To design suitable membership functions (Triangular / Trapezoidal) for fuzzification.
- To construct a fuzzy rule base of at least 6–8 rules relating the inputs to the output.
- To fuzzify a sample input, identify fired rules with their firing strengths, and defuzzify the result using the Weighted Average method to obtain the crisp control output.

3. System Design

The camera autofocus problem is modelled as a Mamdani-type fuzzy inference system (FIS) with two inputs and one output, as described in the following sub-sections.

3.1 Identification of Inputs and Outputs

The controller uses the classical "contrast/sharpness hill-climbing" principle: as the lens moves, an on-sensor sharpness (contrast) value is computed for the image; this value is maximum exactly at the point of correct focus. Two variables derived from this sharpness signal are used as fuzzy inputs, and the required lens motor movement is the fuzzy output.

Variable	Type	Description
Sharpness Error (E)	Input 1	Difference between the current image sharpness value and the sharpness value at the previous lens position; indicates whether focus is improving or degrading.
Rate of Change of Error (dE)	Input 2	Rate at which the sharpness error is changing between successive sampling instants; indicates the trend/speed of change of focus quality.
Lens Motor Step (O)	Output	Magnitude and direction of the stepper-motor drive signal sent to the focusing lens group.

Table 1: Inputs and Output of the Fuzzy Autofocus Controller

3.2 Range of Input and Output Variables

Variable	Range	Unit
Sharpness Error, E	-10 to +10	Normalized contrast units
Rate of Change of Error, dE	-5 to +5	Normalized contrast units / sample

Lens Motor Step, O	-50 to +50	Motor steps (+ = forward, – = backward)

Table 2: Universe of Discourse for Inputs and Output

3.3 Linguistic Variables and Linguistic Values

Each crisp input/output variable is associated with a linguistic variable that takes several linguistic values (fuzzy sets), as summarised in Table 3.

Linguistic Variable	Linguistic Values
Sharpness Error (E)	Negative Large (NL), Negative Small (NS), Zero (Z), Positive Small (PS), Positive Large (PL)
Rate of Change of Error (dE)	Negative (N), Zero (Z), Positive (P)
Lens Motor Step (O)	Negative Large (NL), Negative Small (NS), Zero (Z), Positive Small (PS), Positive Large (PL)

Table 3: Linguistic Variables and Their Linguistic Values

3.4 Membership Functions

Triangular and Trapezoidal membership functions are selected for this design because they are computationally inexpensive (well suited to a real-time embedded camera processor), easy to tune, and adequate for representing the gradual and saturating nature of sharpness error near the extremes of the universe of discourse. The extreme linguistic values (NL and PL) are modelled as trapezoidal sets to represent saturation, while the intermediate values are modelled as triangular sets.

(a) Membership Functions for Sharpness Error (E)

Set	Type	Parameters (a, b, c[, d])
NL	Trapezoidal	(-10, -10, -8, -4)
NS	Triangular	(-8, -4, 0)
Z	Triangular	(-4, 0, 4)

PS	Triangular	(0, 4, 8)
PL	Trapezoidal	(4, 8, 10, 10)

Table 4: Membership Function Parameters for Input E

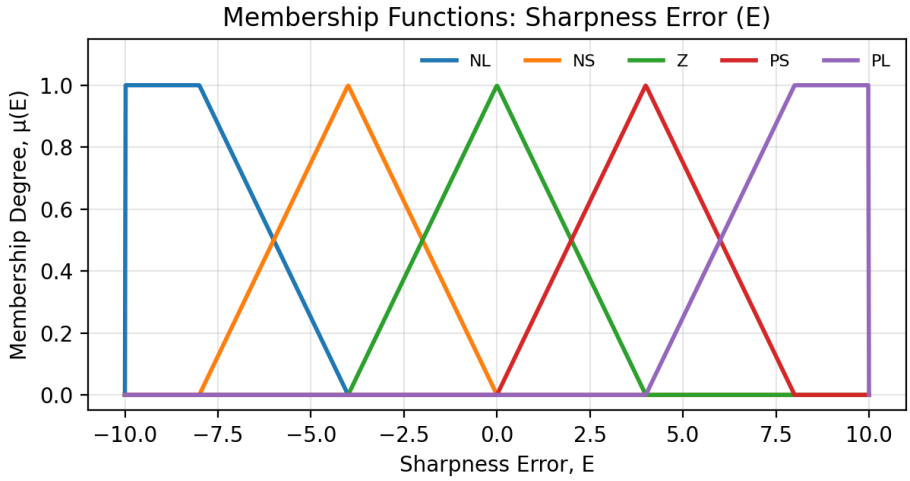


Figure 1: Membership Functions of Sharpness Error (E)

(b) Membership Functions for Rate of Change of Error (dE)

Set	Type	Parameters (a, b, c[, d])
N	Trapezoidal	(-5, -5, -3, 0)
Z	Triangular	(-3, 0, 3)
P	Trapezoidal	(0, 3, 5, 5)

Table 5: Membership Function Parameters for Input dE

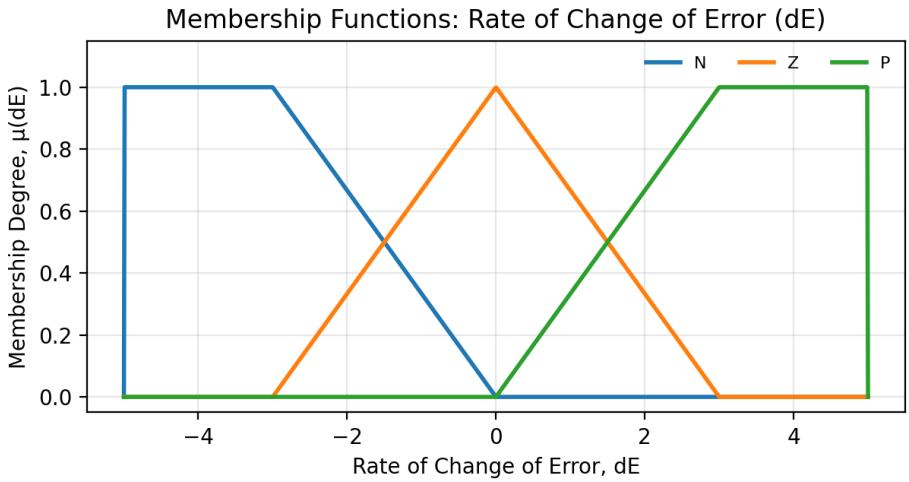


Figure 2: Membership Functions of Rate of Change of Error (dE)

(c) Membership Functions for Lens Motor Step (O)

Set	Type	Parameters (a, b, c[, d])
NL	Trapezoidal	(-50, -50, -40, -20)
NS	Triangular	(-40, -20, 0)
Z	Triangular	(-20, 0, 20)
PS	Triangular	(0, 20, 40)
PL	Trapezoidal	(20, 40, 50, 50)

Table 6: Membership Function Parameters for Output O

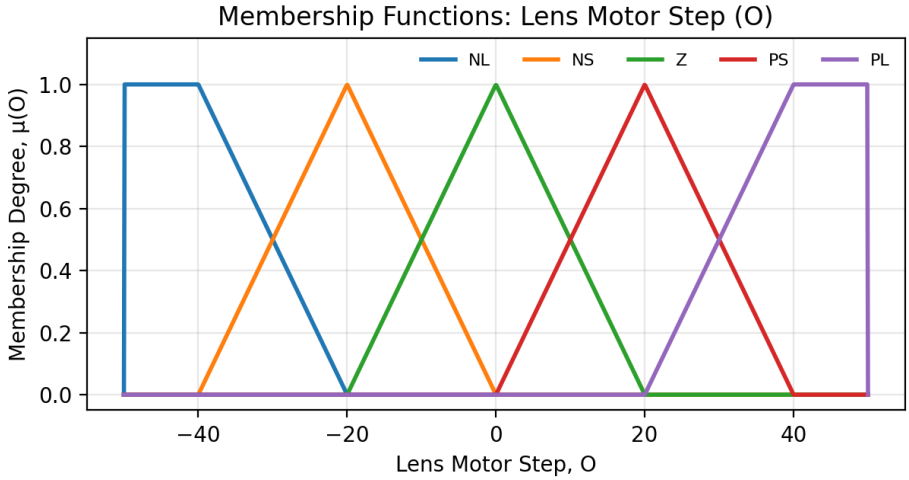


Figure 3: Membership Functions of Lens Motor Step (O)

4. Fuzzy Rule Base

The rule base captures the heuristic control strategy: when the sharpness error is large the lens should move quickly in the direction that reduces the error; when the error is small and steady, only a fine correction (or no correction) is required. The following eight rules were framed based on standard autofocus hill-climbing behaviour.

Rule	IF Sharpness Error (E) is	AND Rate of Change (dE) is	THEN Lens Motor Step (O) is
R1	NL	N	NL
R2	NL	Z	NL
R3	NS	N	NS
R4	NS	Z	NS
R5	Z	Z	Z
R6	PS	Z	PS
R7	PS	P	PL
R8	PL	P	PL

Table 7: Fuzzy Rule Base for the Camera Autofocus Controller

In plain language, Rule R7, for example, reads: “IF the sharpness error is Positive Small AND the rate of change of error is Positive, THEN the lens motor step should be Positive Large” — i.e., focus is degrading further away and worsening quickly, so a large corrective step is issued in time to prevent focus hunting.

5. Sample Data – Fuzzification, Rule Firing and Defuzzification

To illustrate the working of the fuzzy inference engine, one sample data point is taken from the camera's sharpness-detection module and passed through fuzzification, rule evaluation, and weighted-average defuzzification.

5.1 Sample Input Data

- Sharpness Error, E = 4 (normalized contrast units)
- Rate of Change of Error, dE = 2 (normalized contrast units/sample)

5.2 Step 1: Fuzzification – Calculation of Membership Values

For E = 4, using the triangular membership functions of Table 4:

- μZ(4): Z = tri(-4, 0, 4) → at x = 4 (right edge) → μZ(4) = 0
- μPS(4): PS = tri(0, 4, 8) → x = 4 is the peak → μPS(4) = 1.0

All other sets (NL, NS, PL) evaluate to 0 at E = 4, since 4 lies outside their support.

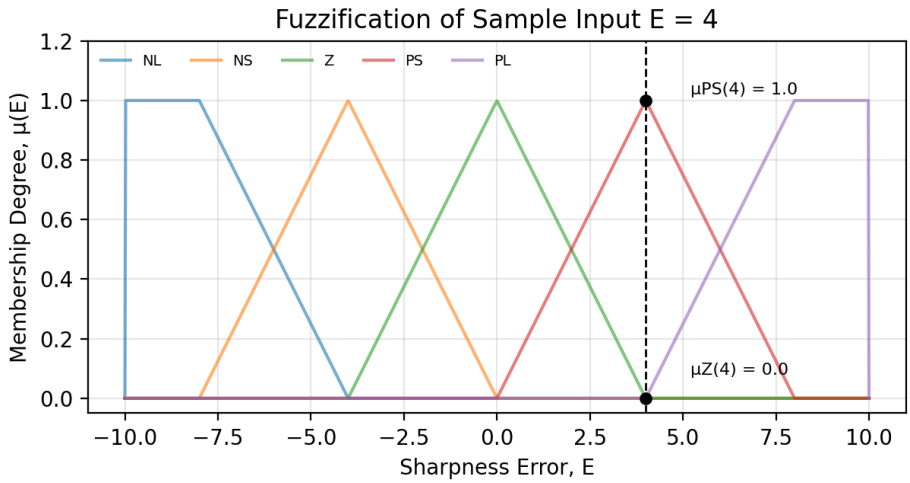


Figure 4: Fuzzification of Sample Input E = 4

For dE = 2, using the membership functions of Table 5:

- μZ(2): Z = tri(-3, 0, 3) → μZ(2) = (3 − 2) / (3 − 0) = 0.333
- μP(2): P = trap(0, 3, 5, 5) → μP(2) = (2 − 0) / (3 − 0) = 0.667

The set N evaluates to 0 at dE = 2.

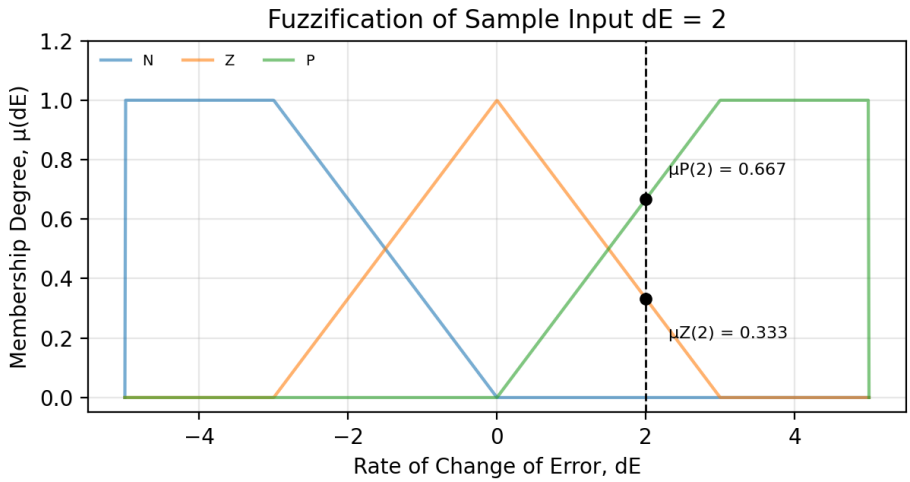


Figure 5: Fuzzification of Sample Input dE = 2

5.3 Step 2: Identification of Fired Rules and Firing Strength

Using the min (AND) operator for rule conjunction, w = min(μE, μdE), each of the eight rules in Table 7 is evaluated. Since μNL(E) = μNS(E) = μZ(E) = 0 and μN(dE) = 0 for this sample, only rules whose antecedents involve E = PS and dE = {Z, P} can fire; all other rules produce zero firing strength and are not listed.

Rule	μE	μdE	Firing Strength w = min(μE, μdE)	Output Set
R6: PS, Z → PS	1.000	0.333	0.333	PS
R7: PS, P → PL	1.000	0.667	0.667	PL

Table 8: Fired Rules and Their Firing Strengths

Thus, two rules are fired for the given sample: Rule R6 with a firing strength of 0.333 (output set PS) and Rule R7 with a firing strength of 0.667 (output set PL). All remaining rules (R1–R5, R8) have zero firing strength for this input and do not contribute to the output.

5.4 Step 3: Defuzzification – Weighted Average Method

The crisp output is computed using the Weighted Average Method, in which the representative (peak) crisp value zi of each fired output set is weighted by its firing strength wi:

$$z^* = \Sigma(w_i \times z_i) / \Sigma(w_i)$$

Using the representative peak values of the output sets (Z = 0, PS = 20, PL = 45 — the mid-point of the trapezoidal plateau for PL) and the firing strengths obtained in Table 8:

Rule	wi	zi (peak value)	wi × zi
R6 (PS)	0.333	20	6.660
R7 (PL)	0.667	45	30.015
Sum	1.000	—	36.675

Table 9: Weighted Average Defuzzification Calculation

$z^* = (0.333 \times 20 + 0.667 \times 45) / (0.333 + 0.667) = 36.675 / 1.000 =$ **36.68**

5.5 Step 4: Crisp Output

The crisp control output obtained is O = 36.68 motor steps (positive direction). This means that for a sharpness error of 4 units which is increasing at a rate of 2 units/sample, the fuzzy controller commands the lens focusing motor to move forward by approximately 37 steps — a large corrective action, which is consistent with the physical expectation that a sizeable and worsening sharpness error requires a fast, large lens movement to restore focus.

6. Result and Conclusion

A two-input, single-output Mamdani fuzzy logic controller was successfully designed for a camera autofocus system. Sharpness Error and its Rate of Change were fuzzified using triangular and trapezoidal membership functions, an eight-rule fuzzy rule base was constructed, and the weighted average method was used to defuzzify the controller output. For the sample input (E = 4, dE = 2), two rules were fired (R6 and R7) and the resulting crisp lens-motor-step command was found to be 36.68 steps in the forward direction. The design demonstrates that fuzzy logic control can provide smooth, human-

like, and computationally light control action for the non-linear and noisy autofocus process, making it well suited for embedded implementation in digital camera systems.

7. References

[1] T. J. Ross, Fuzzy Logic with Engineering Applications, 3rd ed., Wiley, 2010.

[2] L. A. Zadeh, “Fuzzy Sets,” Information and Control, vol. 8, no. 3, pp. 338–353, 1965.

[3] Course/PBL Laboratory Manual: Design of a Fuzzy Controller, Department Lab Manual.

[4] MATLAB Fuzzy Logic Toolbox™ User's Guide, MathWorks.